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## 6 1. Introduction

7 Try to say something [? ].

## 8 2. Reissner–Mindlin plate formulation

### 9 2.1. Governing equations for Mindlin plate

10 Consider a thick plate  $\Omega \times [-\frac{t}{2}, \frac{t}{2}]$  with its mid-surface  $\Omega$  and thickness  $t$  in  
 11 a Cartesian coordinate system, as shown in Figure, where  $x_\alpha$ ,  $\alpha = 1, 2$  are the  
 12 in-plane coordinates on  $\Omega$  and  $x_3$  is the coordinate for the thickness direction.  
 13 In accordance with Reissner–Mindlin hypothesis and inconsideration of the in-  
 14 plane displacement of  $\Omega$ , the displacement  $u_i$ ,  $i = 1, 2, 3$  are given as follows:

$$\begin{cases} u_\alpha(\mathbf{x}) = -x_3\varphi_\alpha(x_1, x_2) \\ u_3(\mathbf{x}) = w(x_1, x_2) \end{cases}, \quad \alpha = 1, 2 \quad (1)$$

15 where  $w$  stands for the deflection and  $\varphi_\alpha$  presents  $\alpha$ th direction component for  
 16 rotation  $\boldsymbol{\varphi}$ . The corresponding infinitesimal strain components  $\varepsilon_{ij}$  are listed as  
 17 follows:

$$\begin{cases} \varepsilon_{\alpha\beta} = \frac{1}{2}(u_{\alpha,\beta} + u_{\beta,\alpha}) = x_3\kappa_{\alpha\beta} \\ \varepsilon_{\alpha 3} = \frac{1}{2}\gamma_\alpha \\ \varepsilon_{33} = 0 \end{cases} \quad (2)$$

18 in which  $\kappa_{\alpha\beta}$  and  $\gamma_\alpha$  are the components for the in-plane curvature  $\boldsymbol{\kappa}$  and shear  
 19 strain  $\boldsymbol{\gamma}$  respectively, the  $\boldsymbol{\kappa}$  and  $\boldsymbol{\gamma}$  are defined as:

$$\boldsymbol{\kappa} = -\frac{1}{2}(\nabla\boldsymbol{\varphi} + \boldsymbol{\varphi}\nabla) \quad (3)$$

$$\boldsymbol{\gamma} = \nabla w - \boldsymbol{\varphi} \quad (4)$$

20 According to the isotropic Mindlin plate theory [1], the stress components  
 21  $\sigma_{ij}$  are given as follows:

$$\begin{cases} \sigma_{\alpha\beta} = D_{\alpha\beta\gamma\eta}\varepsilon_{\gamma\eta} \\ \sigma_{\alpha 3} = G\gamma_\alpha \\ \sigma_{33} = 0 \end{cases} \quad (5)$$

22 where  $D_{\alpha\beta\gamma\eta}$  is denoted as the component for the fourth-order in-plane consti-  
 23 tutive tensor  $\mathbf{D}$  and  $G$  is shear modulus, and they can be evaluated by Young's  
 24 modulus  $E$ , Poisson's ratio  $\nu$  as:

$$\mathbf{D} = \frac{E}{1-\nu^2} \left( \nu \mathbf{1} \otimes \mathbf{1} + \frac{1}{2}(1-\nu)\mathbf{I} \right) \quad (6)$$

$$G = \frac{E}{2(1+\nu)} \quad (7)$$

in which  $\mathbf{1} = \delta_{\alpha\beta} \mathbf{e}_i \otimes \mathbf{e}_j$  and  $\mathbf{I} = \delta_{\alpha\gamma} \delta_{\beta\eta} + \delta_{\alpha\eta} \delta_{\beta\gamma} \mathbf{e}_\alpha \otimes \mathbf{e}_\beta \otimes \mathbf{e}_\gamma \otimes \mathbf{e}_\eta$  represent the second- and fourth-order identity tensors, and  $\delta_{\alpha\beta}$  is Kronecker delta function. Further integrating the stress components through the thickness direction, we can obtain the moment resultant  $\mathbf{m}$  and shear force resultant  $\mathbf{q}$ , given as follows:

$$\mathbf{m} = \int_{-t/2}^{t/2} \sigma_{\alpha\beta} dx_3 \mathbf{e}_\alpha \otimes \mathbf{e}_\beta = \frac{t^3}{12} \mathbf{D} : \boldsymbol{\kappa} \quad (8)$$

$$\mathbf{q} = k \int_{-t/2}^{t/2} \sigma_{\alpha 3} dx_3 \mathbf{e}_\alpha = k G t \boldsymbol{\gamma} \quad (9)$$

where  $k$  presents the shear correction factor that is chosen as  $k = \frac{4}{5}$  in this work. It should be noted that, the geometric parameter for the moment resultant has two more order power of thickness than shear force resultant. As the thickness reducing, the parameter for the moment resultant will much less than that for shear force resultant, for a specific external force, leading to a constraint of shear strain trending to be zero, i.e.  $t \rightarrow 0, \boldsymbol{\gamma} \rightarrow \mathbf{0}$ .

Consequently, the governing equations for Reissner-Mindlin plate can be expressed as follows:

$$\begin{cases} \nabla \cdot \mathbf{m} - \mathbf{q} = \mathbf{0} & \text{in } \Omega \\ \nabla \cdot \mathbf{q} + \bar{p} = 0 & \text{in } \Omega \\ \mathbf{q} \cdot \mathbf{n} = \bar{q}_n & \text{on } \Gamma_q \\ \mathbf{m} \cdot \mathbf{n} = \bar{\mathbf{m}}_n & \text{on } \Gamma_m \\ \varphi = \bar{\varphi} & \text{on } \Gamma_\varphi \\ w = \bar{w} & \text{on } \Gamma_w \end{cases} \quad (10)$$

where  $\bar{p}$  is the transverse load,  $\bar{q}_n$  and  $\bar{\mathbf{m}}$  are the prescribed shear force and moment on the natural boundary  $\Gamma_q$  and  $\Gamma_m$  respectively,  $\bar{w}$  and  $\bar{\varphi}$  are the prescribed deflection and rotation on the essential boundary  $\Gamma_w$  and  $\Gamma_\varphi$  respectively, and  $\mathbf{n}$  is the unit outward normal vector on the boundary. All the boundary segments satisfy the following relationship:

$$\Gamma_q \cup \Gamma_m = \Gamma_w \cup \Gamma_\varphi = \Gamma, \quad \Gamma_w \cap \Gamma_\varphi = \Gamma_q \cap \Gamma_m = \emptyset \quad (11)$$

and  $\Gamma$  is the boundary of  $\Omega$ .

## 2.2. Mixed-formulation

The thick plate Galerkin formulation generally use the three-variable weak form to immigrate the locking issue. To obtain the corresponding Galerkin weak form, the complementary energy functional is given by:

$$\begin{aligned} \Pi(\mathbf{u}, \mathbf{q}) = & \int_{\Omega} \frac{t^3}{24} \boldsymbol{\kappa} : \mathbf{D} : \boldsymbol{\kappa} d\Omega - \int_{\Omega} \frac{1}{2kGt} \mathbf{q} \cdot \mathbf{q} d\Omega - \int_{\Omega} \varphi \cdot \mathbf{q} d\Omega \\ & + \int_{\Gamma_w} \mathbf{q} \cdot \mathbf{n} \bar{w} d\Gamma + \int_{\Gamma_m} \varphi \cdot \bar{\mathbf{m}}_n d\Gamma + \int_{\Gamma_q} w(\mathbf{q} \cdot \mathbf{n} - \bar{q}_n) d\Gamma \\ & - \int_{\Omega} w(\nabla \cdot \mathbf{q} + \bar{p}) d\Omega \end{aligned} \quad (12)$$

47 where  $\mathbf{u}$  stands for a displacement set containing deflection  $w$  and rotation  $\boldsymbol{\varphi}$ ,  
 48 i.e.  $\mathbf{u} = \{w, \boldsymbol{\varphi}\}$ . Introducing the standard variational operation herein leads to  
 49 the following Galerkin weak form: find  $w \in W$ ,  $\boldsymbol{\varphi} \in \Phi$ ,  $\mathbf{q} \in Q$ ,

$$\begin{aligned} a_\varphi(\delta\boldsymbol{\varphi}, \boldsymbol{\varphi}) + b_\varphi(\delta\boldsymbol{\varphi}, \mathbf{q}) &= f_\varphi(\delta\boldsymbol{\varphi}), \quad \forall \delta\boldsymbol{\varphi} \in \Phi \\ b_w(\delta w, \mathbf{q}) &= f_w(\delta w), \quad \forall \delta w \in W \\ a_q(\delta\mathbf{q}, \mathbf{q}) + b_q(\delta\mathbf{q}, \mathbf{u}) &= f_q(\delta\mathbf{q}), \quad \forall \delta\mathbf{q} \in Q \end{aligned} \quad (13)$$

50 where the deflection space  $W$ , rotation space  $\Phi$  and shear force space  $Q$  are  
 51 defined as:

$$W := \{\} \quad (14)$$

$$\Phi := \{\} \quad (15)$$

$$Q := \{\} \quad (16)$$

52 and the bilinear forms and linear forms in Eq. (13) are detailed as follows:

$$a_\varphi(\delta\boldsymbol{\varphi}, \boldsymbol{\varphi}) = \int_{\Omega} \delta\boldsymbol{\kappa} : \frac{t^3}{12} \mathbf{D} : \boldsymbol{\kappa} d\Omega \quad (17)$$

$$a_q(\delta\mathbf{q}, \mathbf{q}) = - \int_{\Omega} \frac{1}{2kGt} \delta\mathbf{q} \cdot \mathbf{q} d\Omega \quad (18)$$

$$b_q(\mathbf{q}, \mathbf{u}) = b_w(w, \mathbf{q}) + b_\varphi(\boldsymbol{\varphi}, \mathbf{q}) \quad (19)$$

$$b_\varphi(\boldsymbol{\varphi}, \mathbf{q}) = - \int_{\Omega} \boldsymbol{\varphi} \cdot \mathbf{q} d\Omega \quad (20)$$

$$\begin{aligned} b_w(w, \mathbf{q}) &= \int_{\Gamma_q} w \mathbf{q} \cdot \mathbf{n} d\Gamma - \int_{\Omega} w \nabla \cdot \mathbf{q} d\Omega \\ &= \int_{\Omega} \nabla w \cdot \mathbf{q} d\Omega - \int_{\Gamma_w} w \mathbf{q} \cdot \mathbf{n} d\Gamma \end{aligned} \quad (21)$$

$$f_\varphi(\boldsymbol{\varphi}) = - \int_{\Gamma_m} \boldsymbol{\varphi} \cdot \bar{\mathbf{m}}_n d\Gamma \quad (22)$$

$$f_w(w) = \int_{\Gamma_q} w \bar{q}_n d\Gamma + \int_{\Omega} w \bar{p} d\Omega \quad (23)$$

$$f_q(\mathbf{q}) = - \int_{\Gamma_w} \mathbf{q} \cdot \mathbf{n} \bar{w} d\Gamma \quad (24)$$

53 Furthermore, we introduce the approximation schemes to discrete these three  
 54 variables, namely  $w_h$ ,  $\boldsymbol{\varphi}_h$ ,  $\mathbf{q}_h$ , these approximants are represented by shape  
 55 functions  $N_I^w$ ,  $N_K^\varphi$ ,  $N_M^q$  and their corresponding nodal coefficients  $w_I$ ,  $\boldsymbol{\varphi}_K$ ,  $\mathbf{q}_M$

56 respectively, yields:

$$\begin{cases} w_h(\mathbf{x}) = \sum_{I=1}^{n_w} N_I^w(\mathbf{x}) w_I \\ \varphi_h(\mathbf{x}) = \sum_{K=1}^{n_\varphi} N_K^\varphi \varphi_K \\ \mathbf{q}_h(\mathbf{x}) = \sum_{R=1}^{n_q} N_R^q(\mathbf{x}) \mathbf{q}_R \end{cases} \quad (25)$$

57 Substituting Eq. (25) back into Eq. (13) can obtain the Galerkin weak formu-  
58 lation: find  $w_h \in W_h$ ,  $\varphi_h \in \Phi_h$ ,  $\mathbf{q}_h \in Q_h$ ,

$$a(\delta\varphi_h, \varphi_h) + b_\varphi(\delta\varphi_h, \mathbf{q}_h) = f_\varphi(\delta\varphi_h), \quad \forall \delta\varphi_h \in \Phi_h \quad (26a)$$

$$b_w(\delta w_h, \mathbf{q}_h) = f_w(\delta w_h), \quad \forall \delta w_h \in W_h \quad (26b)$$

$$b(\delta\mathbf{q}_h, \mathbf{u}_h) = f_q(\delta\mathbf{q}_h), \quad \forall \delta\mathbf{q}_h \in Q_h \quad (26c)$$

59 where  $\mathbf{u}_h = \{w_h, \varphi_h\}^T$ . The related approximation spaces are defined as:

$$W_h := \text{span}\{N_I^w\}_{I=1}^{n_w} \quad (27)$$

$$\Phi_h := \text{span}\{N_K^\varphi\}_{K=1}^{n_\varphi} \quad (28)$$

$$Q_h := \text{span}\{N_R^q\}_{R=1}^{n_q} \quad (29)$$

60 FIXME: approximation types

$$\begin{bmatrix} \mathbf{K}^{\varphi\varphi} & \mathbf{0} & \mathbf{K}^{q\varphi T} \\ \mathbf{0} & \mathbf{0} & \mathbf{K}^{qwT} \\ \mathbf{K}^{q\varphi} & \mathbf{K}^{qw} & \mathbf{K}^{qq} \end{bmatrix} \begin{Bmatrix} \mathbf{d}^\varphi \\ \mathbf{d}^w \\ \mathbf{d}^q \end{Bmatrix} = \begin{Bmatrix} \mathbf{f}^\varphi \\ \mathbf{f}^w \\ \mathbf{f}^q \end{Bmatrix} \quad (30)$$

$$\left( \underbrace{\begin{bmatrix} \mathbf{K}^{\varphi\varphi} & \mathbf{0} \\ \mathbf{0} & \mathbf{0} \end{bmatrix}}_{\mathbf{K}^{\text{bending}}} + \underbrace{\begin{bmatrix} \tilde{\mathbf{K}}^{\varphi\varphi} & \tilde{\mathbf{K}}^{\varphi w} \\ \tilde{\mathbf{K}}^{w\varphi} & \tilde{\mathbf{K}}^{ww} \end{bmatrix}}_{\tilde{\mathbf{K}}^{\text{shearing}}} \right) \begin{Bmatrix} \mathbf{d}^\varphi \\ \mathbf{d}^w \end{Bmatrix} = \begin{Bmatrix} \mathbf{f}^\varphi - \tilde{\mathbf{f}}^\varphi \\ \mathbf{f}^w - \tilde{\mathbf{f}}^w \end{Bmatrix} \quad (31)$$

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$$\tilde{\mathbf{K}}^{\varphi\varphi} = -\mathbf{K}^{q\varphi T} (\mathbf{K}^{qq})^{-1} \mathbf{K}^{q\varphi} \quad (32)$$

$$\tilde{\mathbf{K}}^{\varphi w} = -\mathbf{K}^{q\varphi T} (\mathbf{K}^{qq})^{-1} \mathbf{K}^{qw} \quad (33)$$

$$\tilde{\mathbf{K}}^{ww} = -\mathbf{K}^{qwT} (\mathbf{K}^{qq})^{-1} \mathbf{K}^{qw} \quad (34)$$

62 **Remark 1.** As shown in Eq. (26), there are two constraints in thick plate  
63 mixed formulation. One, as listed in Eq. (26b), is a constraint mainly regarding  
64 the shear force resultant on natural boundary  $\Gamma_q$  and its divergence in entire  
65 domain. The other is deflection essential boundary condition  $\Gamma_w$  and the shear  
66 strain enforcement in whole domain, as listed in Eq. (26c). However, for the  
67 most of pioneer works, only consider the shear strain constraint, and that is not  
68 sufficient for a satisfaction result.

69 **Remark 2.**

### 70 3. Optimal constraint ratio for shear locking

71 The previous section has been mentioned that the shear locking is consisted  
 72 of two parts, shear force locking and shear strain locking. The corresponding  
 73 LBB conditions are listed as follows:

- 74 • Shear force locking LBB condition:

$$\inf_{w_h \in W_h} \sup_{\mathbf{q}_h \in Q_h} \frac{|b_w(w_h, \mathbf{q}_h)|}{\|w_h\|_W \|\mathbf{q}_h\|_Q} \geq \beta_q \quad (35)$$

- 75 • Shear strain locking LBB condition:

$$\inf_{\mathbf{q}_h \in Q_h} \sup_{\mathbf{u}_h \in W_h \times \Phi_h} \frac{|b(\mathbf{q}_h, \mathbf{u}_h)|}{\|\mathbf{q}_h\|_Q \|\mathbf{u}_h\|_{W \times \Phi}} \geq \beta_\gamma \quad (36)$$

76 where,  $\beta_q$  and  $\beta_\gamma$  are LBB coefficients. The norms in the above LBB conditions  
 77 are defined as follows:

$$\|w_h\|_W^2 = \int_{\Omega} \delta w_h k G h w_h d\Omega \quad (37)$$

$$\|\mathbf{q}_h\|_Q^2 = \int_{\Omega} \frac{\delta \mathbf{q}_h \cdot \mathbf{q}_h}{k G t} d\Omega \quad (38)$$

$$\|\boldsymbol{\varphi}_h\|_{\Phi}^2 = \int_{\Omega} \delta \boldsymbol{\kappa}_h : \frac{t^3}{12} \mathbf{D} : \boldsymbol{\kappa}_h d\Omega \quad (39)$$

$$\|\mathbf{u}_h\|_{W \times \Phi}^2 = \|w_h\|_W^2 + \|\boldsymbol{\varphi}_h\|_{\Phi}^2 \quad (40)$$

78 Next, we will focus on these two LBB conditions respectively and employ the  
 79 complete-polynomial-based estimate method [2] to investigate the relationship  
 80 between their LBB coefficients  $\beta_q$  and  $\beta_\gamma$  and the degree of freedoms (DOFs)  
 81 for variables  $w_h$ ,  $\boldsymbol{\varphi}_h$  and  $\mathbf{q}_h$ . The detailed path for the complete-polynomial-  
 82 based estimate method is listed in Appendix A. Consequently, the optimal ratio  
 83 of degrees of freedoms (DOFs) between variables can be revealed based on these  
 84 two estimators.

#### 85 3.1. Estimator for $\beta_q$

86 The coefficient  $\beta_q$  controls the behavior of shear force locking in Eq. (26b),  
 87 the detailed enforcement is listed as follows:

$$\begin{aligned} & b_w(\delta w_h, \mathbf{q}_h) - f_w(\delta w_h) = (\delta w_h, \mathcal{W} \mathbf{q}_h) - f_w(\delta w_h) \\ & = \int_{\Gamma_q} \delta w_h (\mathbf{q}_h \cdot \mathbf{n} - \bar{q}_n) d\Gamma - \int_{\Omega} \delta w_h (\nabla \cdot \mathbf{q}_h + \bar{p}) d\Omega \\ & = 0 \end{aligned} \quad (41)$$

88 where, the constraints includes two parts, one is the equilibrium equation be-  
 89 tween shear force resultant and transverse load in  $\Omega$ , the other is the natrual

90 shear force boundary condition on  $\Gamma_q$ . In accordance with the general form of  
 91 completed-polynomial-based estimator mentioned in Appendix A, the operator  
 92  $\mathcal{W} : Q \rightarrow W$  is a specific instance of  $\mathcal{P}$  in Theorem.1, and is defined as:

$$\mathcal{W} = \begin{cases} \mathcal{W}^\Omega = -\nabla \cdot & \text{in } \Omega \\ \mathcal{W}^{\Gamma_q} \mathcal{T}^{\Gamma_q}, \mathcal{W}^{\Gamma_q} = \mathbf{n} \cdot & \text{on } \Gamma_q \end{cases} \quad (42)$$

93 where, based on the trace theorem [],  $\mathcal{T}^{\Gamma_q} : Q \rightarrow Q|_{\Gamma_q}$  is a trace operator map-  
 94 ping the shear force  $\mathbf{q}$  from domain  $\Omega$  to the boundary  $\Gamma_q$ , while the parametric  
 95 variable of  $\mathbf{q}$  on boundary can be represented using  $s$ , i.e.  $\mathbf{q}(s) \in Q|_{\Gamma_q}$ . Ac-  
 96 cording to Theorem.1, the completed-polynomial-based estimator of  $\beta_q$  has the  
 97 following form:

$$\beta_q \leq \inf_{Q' \subset Q_{n_q} \setminus \ker \mathcal{W}_h \mathcal{I}_h^q} \sup_{\mathbf{q} \in Q'} \frac{\|\mathcal{W}\mathbf{q}\|_W}{\|\mathbf{q}\|_Q} + O(h) \quad (43)$$

98 in which  $\mathcal{I}_h^q : Q_{n_q} \rightarrow Q_h$  is the interpolation operator for shear force resultant.  
 99  $\mathcal{W}_h : Q_h \rightarrow W_h$  is the orthogonal operator of  $\mathcal{W}$ , i.e.  $(\delta w_h, \mathcal{W}\mathbf{q}_h - \mathcal{W}_h\mathbf{q}_h) = 0$ .  
 100  $Q_{n_q} \subset Q$  is the polynomial space that has the identical dimensions with  $Q_h$ , i.e.  
 101  $\dim Q_{n_q} = \dim Q_h = 2n_q$ .

102 From Eq. (43), the LBB condition of Eq. (35) can be maintained while the  
 103 first term on right-hand side of Eq. (43) is non-zero, which should be necessarily  
 104 required that:

$$\dim(Q_{n_q} \setminus \ker \mathcal{W}_h \mathcal{I}_h^q) \leq \dim(Q_{n_q} \setminus \ker \mathcal{W}) \quad (44)$$

105 where, obviously, leading the definitions of  $\mathcal{I}_h^q$  and  $Q_{n_q}$ , and the rank-nullity  
 106 theorem, the left-hand side of Eq. (44) can be rewritten as:

$$\begin{aligned} \dim(Q_{n_q} \setminus \ker \mathcal{W}_h \mathcal{I}_h^q) &= \dim(Q_h \setminus \ker \mathcal{W}_h) \\ &= \dim Q_h - \dim(\ker \mathcal{W}_h) \\ &= \dim(\text{Im} \mathcal{W}_h) \\ &= \dim W_h = n_w \end{aligned} \quad (45)$$

107 In contrast, to evaluate the right-hand side of Eq. (44), we first introduce  
 108 the definition of operator  $\mathcal{W}$  in (42),  $\ker \mathcal{W}$  can be represented as:

$$\ker \mathcal{W} = \ker \mathcal{W}^\Omega \cap \ker \mathcal{W}^{\Gamma_q} \mathcal{T}^{\Gamma_q} \quad (46)$$

109 According to above equation, the space  $Q_{n_q} \setminus \ker \mathcal{W}$  can be expressed by two  
 110 terms, one is regarding the kernel space of  $\mathcal{W}^\Omega$ , the other is regarding the kernel  
 111 space of  $\mathcal{W}^{\Gamma_q} \mathcal{T}^{\Gamma_q}$ , yields:

$$\begin{aligned} \dim(Q_{n_q} \setminus \ker \mathcal{W}) &= \dim(Q_{n_q} \setminus (\ker \mathcal{W}^\Omega \cap \ker \mathcal{W}^{\Gamma_q} \mathcal{T}^{\Gamma_q})) \\ &= \dim(Q_{n_q} \setminus \ker \mathcal{W}^\Omega) \cup \dim(Q_{n_q} \setminus \ker \mathcal{W}^{\Gamma_q} \mathcal{T}^{\Gamma_q}) \\ &= \dim(Q_{n_q} \setminus \ker \mathcal{W}^\Omega) + \dim(Q_{n_q} \setminus \ker \mathcal{W}^{\Gamma_q} \mathcal{T}^{\Gamma_q}) \\ &\quad - \dim((Q_{n_q} \setminus \ker \mathcal{W}^\Omega) \cap (Q_{n_q} \setminus \ker \mathcal{W}^{\Gamma_q} \mathcal{T}^{\Gamma_q})) \end{aligned} \quad (47)$$

where the second and third equations of Eq. (47) is derived based on De Morgan's theorem[] and Grassmann's theorem[] respectively. For clarity, these two theorems are illustrated in Appendix B.

For the first term regarding the kernel space of  $\mathcal{W}^\Omega$ , in accordance with rank-nullity theorem[3], the dimension of this space is equal to the that of the image space of  $\mathcal{W}^\Omega$  as:

$$\dim(Q_{n_q} \setminus \ker \mathcal{W}^\Omega) = \dim Q_{n_q} - \dim(\ker \mathcal{W}^\Omega) = \dim(\text{Im} \mathcal{W}^\Omega) \quad (48)$$

Because of the  $Q_{n_q}$  is the second order polynomial tensor space with dimension of  $2n_q$  and the  $\mathcal{W}^\Omega$  is the divergence operator acting on  $Q_{n_q}$ , the polynomial in  $\text{Im} \mathcal{W}^\Omega$  is one order lower than polynomial order of  $Q_{n_q}$ . The following equation presents the detailed monomial basis of  $Q_{n_q}$ , and the kernel space of operator  $\mathcal{W}^\Omega$ :

$$Q_{n_q} = \text{span} \left\{ \underbrace{\begin{pmatrix} 1 \\ 0 \end{pmatrix}, \begin{pmatrix} 0 \\ 1 \end{pmatrix}, \begin{pmatrix} y \\ 0 \end{pmatrix}, \begin{pmatrix} 0 \\ x \end{pmatrix}, \begin{pmatrix} x \\ -y \end{pmatrix}, \dots, \begin{pmatrix} x \\ y \end{pmatrix}, \dots}_{\ker \mathcal{W}^\Omega} \right\} \quad (49)$$

To simplify the expression, we define two operators to implement the transformation between the order of a complete polynomial space, namely  $n_o$  and its monomial basis number, namely  $n_m$ :

$$n_o = [n_p] = \frac{\sqrt{1+8n_p}-3}{2}, \quad n_p = \langle n_o \rangle = \frac{(n_o+1)(n_o+2)}{2} \quad (50)$$

where  $\langle [n] \rangle = n$ . Therefore, the dimension of  $\text{Im} \mathcal{W}^\Omega$  can be estimated by:

$$\langle \lfloor [n_q] \rfloor - 1 \rangle \leq \dim(\text{Im} \mathcal{W}^\Omega) \leq \langle \lceil [n_q] \rceil - 1 \rangle \quad (51)$$

in which  $\lfloor \bullet \rfloor$  and  $\lceil \bullet \rceil$  are the floor and ceiling functions, respectively.

For the second term regarding the kernel space of  $\mathcal{W}^{\Gamma_q} \mathcal{T}^{\Gamma_q}$ , leading the Lemma.1 listed in Appendix B, the dimension of  $Q_{n_q} \setminus \ker \mathcal{W}^{\Gamma_q} \mathcal{T}^{\Gamma_q}$  becomes:

$$\begin{aligned} \dim(Q_{n_q} \setminus \ker \mathcal{W}^{\Gamma_q} \mathcal{T}^{\Gamma_q}) &= \dim(Q_{n_q^\Gamma} \setminus \ker \mathcal{W}^{\Gamma_q}) \\ &= \dim Q_{n_q^\Gamma} - \dim(\ker \mathcal{W}^{\Gamma_q}) \\ &= \dim(\text{Im} \mathcal{W}^{\Gamma_q}) \end{aligned} \quad (52)$$

where  $Q_{n_q^\Gamma} \subset Q|_{\Gamma_q}$  is the polynomial space with the identical dimension of trace space  $\mathcal{T}^{\Gamma_q}(Q_h)$ , i.e.  $\dim Q_{n_q^\Gamma} = \dim \mathcal{T}^{\Gamma_q}(Q_h) = 2n_q^\Gamma$ . For instance, the detailed monomial basis of  $Q_{n_q^\Gamma}$  is listed as follows:

$$Q_{n_q^\Gamma} = \text{span} \left\{ \underbrace{\begin{pmatrix} n_2 \\ -n_1 \end{pmatrix}, \begin{pmatrix} n_2 s \\ -n_1 s \end{pmatrix}, \begin{pmatrix} n_2 s^2 \\ -n_1 s^2 \end{pmatrix}, \dots}_{\ker \mathcal{W}^{\Gamma_q}}, \underbrace{\begin{pmatrix} n_2 \\ n_1 \end{pmatrix}, \begin{pmatrix} n_2 s \\ n_1 s \end{pmatrix}, \begin{pmatrix} n_2 s^2 \\ n_1 s^2 \end{pmatrix}, \dots}_{Q_{n_q^\Gamma} \setminus \ker \mathcal{W}^{\Gamma_q}} \right\} \quad (53)$$



133 It is clear that, from Eq. (53), the dimension of second term is evaluated as:

$$\dim(Q_{n_q} \setminus \ker \mathcal{W}^{\Gamma_q} \mathcal{T}^{\Gamma_q}) = n_q^\Gamma \quad (54)$$

134 With the consideration of Eqs. (47), (51) and (54), the right-hand side of  
135 Eq. (44) becomes to:

$$\max(\langle \lfloor [n_q] \rfloor - 1 \rangle, n_q^\Gamma) \leq \dim(Q_{n_q} \setminus \ker \mathcal{W}) \leq \langle \lceil [n_q] \rceil - 1 \rangle + n_q^\Gamma \quad (55)$$

136 It is noted that the dimension of the trace space  $\mathcal{T}^{\Gamma_q}(Q_h)$  is generally iden-  
137 tical to the number of DOFs on  $\Gamma_q$ , i.g. for finite element approximation. So  
138 generally speaking,  $n_q^\Gamma \ll \langle \lfloor [n_q] \rfloor - 1 \rangle$ . To totally ensure the LBB condition of  
139 Eq. (35), considering Eqs. (44), (45) and (55), the  $n_w$  should be always less  
140 than the low bound of  $Q_{n_q} \setminus \ker \mathcal{W}$ 's dimension, i.e.:

$$n_w \leq \langle \lfloor [n_q] \rfloor - 1 \rangle \quad (56)$$

### 141 3.2. Estimator for $\beta_\gamma$

142 The behavior of shear strain locking in Eq. (36) is controlled by the coeffi-  
143 cient  $\beta_\gamma$ , where the shear strain locking contains two parts as follows:

$$\begin{aligned} & b(\delta \mathbf{q}_h, \mathbf{u}_h) - f_q(\delta \mathbf{q}_h) = (\delta \mathbf{q}_h, \mathcal{Q} \mathbf{u}_h) - f_q(\delta \mathbf{q}_h) \\ & = \int_{\Omega} \delta \mathbf{q} \cdot (\nabla w - \boldsymbol{\varphi}) d\Omega - \int_{\Gamma_w} \delta \mathbf{q} \cdot \mathbf{n} (w - \bar{w}) d\Gamma \\ & = 0 \end{aligned} \quad (57)$$

144 where the product operator herein stands for the tensors. And the operator  
145  $\mathcal{Q} : W \times \Phi \rightarrow Q$  is defined individually in the domain  $\Omega$  and on the boundary  
146  $\Gamma_w$  by:

$$\mathcal{Q} = \begin{cases} \mathcal{Q}^\Omega = \{\nabla, -\} & \text{in } \Omega \\ \mathcal{Q}^{\Gamma_w} \mathcal{T}^w, \mathcal{Q}^{\Gamma_w} = -\mathbf{n} & \text{on } \Gamma_w \end{cases} \quad (58)$$

147 in which, in accordance with trace theorem [],  $\mathcal{T}^w : W \times \Phi \rightarrow W|_{\Gamma_w}$  is the trace  
148 operator mapping displacement  $\mathbf{u}$  from domain  $\Omega$  to boundary  $\Gamma_w$ . Again,  
149 introducing the Theorem. 1, the  $\beta_\gamma$  can further be bounded by:

$$\beta_\gamma \leq \inf_{U' \subset W_{n_w} \times \Phi_{n_\varphi} \setminus \ker \mathcal{Q}_h \mathcal{I}_h^u} \sup_{\mathbf{u}_h \in U'} \frac{\|\mathcal{Q} \mathbf{u}\|_Q}{\|\mathbf{u}\|_{W \times \Phi}} + O(h) \quad (59)$$

150 where  $\mathcal{I}_h^u : W_{n_w} \times \Phi_{n_\varphi} \rightarrow W_h \times \Phi_h$  is the interpolation operator for deflection  
151  $w$  and rotation  $\boldsymbol{\varphi}$ , and  $W_{n_w}$ ,  $\Phi_{n_\varphi}$  are the polynomial spaces that have the  
152 identical dimensions with  $W_h$ ,  $\Phi_h$  respectively, i.e.  $\dim W_{n_w} = \dim W_h = n_w$ ,  
153  $\dim \Phi_{n_\varphi} = \dim \Phi_h = 2n_\varphi$ .

154 To maintain the LBB condition of (36), i.e.  $\beta_\gamma$  is a coefficient independent  
155 with characterized length  $h$ , the dimensions of the relative spaces in Eq. (59)  
156 should satisfy the following relationship:

$$\dim(W_{n_w} \times \Phi_{n_\varphi} \setminus \ker \mathcal{Q}_h \mathcal{I}_h^u) \leq \dim(W_{n_w} \times \Phi_{n_\varphi} \setminus \ker \mathcal{Q}) \quad (60)$$

157 The left-hand side of Eq. (60) can be represented by the DOFs of approximated  
 158 shear force resultant:

$$\begin{aligned}
 \dim(W_{n_w} \times \Phi_{n_\varphi} \setminus \ker \mathcal{Q}_h \mathcal{I}_h^u) &= \dim(W_h \times \Phi_h \setminus \ker \mathcal{Q}_h) \\
 &= \dim(W_h \times \Phi_h) - \dim(\ker \mathcal{Q}_h) \\
 &= \dim(\text{Im } \mathcal{Q}_h) \\
 &= \dim \mathcal{Q}_h = 2n_q
 \end{aligned} \tag{61}$$

159 On the other hand, with De Morgan's theorem and Grassmann's theorem [],  
 160 the right-hand side of Eq. (60) becomes:

$$\begin{aligned}
 &\dim(W_{n_w} \times \Phi_{n_\varphi} \setminus \ker \mathcal{Q}) \\
 &= \dim(W_{n_w} \times \Phi_{n_\varphi} \setminus (\ker \mathcal{Q}^\Omega \cap \ker \mathcal{Q}^{\Gamma_w} \mathcal{T}^{\Gamma_w})) \\
 &= \dim(W_{n_w} \times \Phi_{n_\varphi} \setminus \ker \mathcal{Q}^\Omega) \cup \dim(W_{n_w} \times \Phi_{n_\varphi} \setminus \ker \mathcal{Q}^{\Gamma_w} \mathcal{T}^{\Gamma_w}) \\
 &= \dim(W_{n_w} \times \Phi_{n_\varphi} \setminus \ker \mathcal{Q}^\Omega) + \dim(W_{n_w} \times \Phi_{n_\varphi} \setminus \ker \mathcal{Q}^{\Gamma_w} \mathcal{T}^{\Gamma_w}) \\
 &\quad - \dim((W_{n_w} \times \Phi_{n_\varphi} \setminus \ker \mathcal{Q}^\Omega) \cap (W_{n_w} \times \Phi_{n_\varphi} \setminus \ker \mathcal{Q}^{\Gamma_w} \mathcal{T}^{\Gamma_w}))
 \end{aligned} \tag{62}$$

161 where the relationship of  $\ker \mathcal{Q} = \ker \mathcal{Q}^\Omega \cap \ker \mathcal{Q}^{\Gamma_w} \mathcal{T}^{\Gamma_w}$  is used herein. Similarly,  
 162 this dimension of Eq. (62) is consisted of two terms, one relates to the kernel  
 163 space of  $\mathcal{Q}^\Omega$  in  $\Omega$ , the other relates to the kernel space of  $\mathcal{Q}^{\Gamma_w} \mathcal{T}^{\Gamma_w}$  on  $\Gamma_w$ .

164 The dimension of first term regarding the kernel space of  $\mathcal{Q}^\Omega$  can be revealed  
 165 by the detailed expression of monomial basis of  $W_{n_w} \times \Phi_{n_\varphi}$  as:

$$W_{n_w} \times \Phi_{n_\varphi} = \text{span} \left\{ \overbrace{\left( \begin{pmatrix} 1 \\ 0 \\ 0 \end{pmatrix}, \begin{pmatrix} x \\ 1 \\ 0 \end{pmatrix}, \begin{pmatrix} y \\ 0 \\ 1 \end{pmatrix}, \begin{pmatrix} x^2 \\ 2x \\ 0 \end{pmatrix}, \begin{pmatrix} xy \\ y \\ x \end{pmatrix}, \begin{pmatrix} y^2 \\ 0 \\ 2y \end{pmatrix}, \dots \right)}^{\ker \mathcal{Q}^\Omega}, \underbrace{\left( \begin{pmatrix} 0 \\ 1 \\ 0 \end{pmatrix}, \begin{pmatrix} 0 \\ 0 \\ 1 \end{pmatrix}, \begin{pmatrix} 0 \\ x \\ 0 \end{pmatrix}, \begin{pmatrix} 0 \\ 0 \\ x \end{pmatrix}, \begin{pmatrix} 0 \\ y \\ 0 \end{pmatrix}, \begin{pmatrix} 0 \\ 0 \\ y \end{pmatrix}, \dots \right)}_{W_{n_w} \times \Phi_{n_\varphi} \setminus \ker \mathcal{Q}^\Omega} \right\} \tag{63}$$

166 Obviously, the monomial basis of deflection  $w$  is one order higher than that of  
 167 rotation in  $\ker \mathcal{Q}^\Omega$ . So the dimension of  $\ker \mathcal{Q}^\Omega$  can be evaluated by:

$$\dim \ker \mathcal{Q}^\Omega = \min(n_w, n_s) \tag{64}$$

168 with

$$\langle \lfloor n_\varphi \rfloor + 1 \rangle \leq n_s \leq \langle \lceil n_\varphi \rceil + 1 \rangle \tag{65}$$

169 where  $n_w$  presents the number of monomial basis for  $w$ .  $n_s$  stands for the  
 170 monomial basis number with one order higher than  $\Phi_{n_\varphi}$ 's monomial basis. Then,  
 171 the dimension of  $W_{n_w} \times \Phi_{n_\varphi} \setminus \ker \mathcal{Q}^\Omega$  can be evaluated by:

$$\begin{aligned}
 \dim(W_{n_w} \times \Phi_{n_\varphi} \setminus \ker \mathcal{Q}^\Omega) &= \dim(W_{n_w} \times \Phi_{n_\varphi}) - \dim(\ker \mathcal{Q}^\Omega) \\
 &= n_w + 2n_\varphi - \min(n_w, n_s)
 \end{aligned} \tag{66}$$

172 To further calculate the dimension of the second term regarding the kernel  
 173 space of  $\mathcal{Q}^{\Gamma_w} \mathcal{T}^{\Gamma_w}$ , we first suppose the trace space  $\mathcal{T}^{\Gamma_w}(W_h \times \Phi_h)$  and its  
 174 corresponding polynomial space  $W_{n_w}^{\Gamma}$ ,  $\dim W_{n_w}^{\Gamma} = \dim \mathcal{T}^{\Gamma_w}(W_h \times \Phi_h) = n_w^{\Gamma}$ .  
 175 Using the Lemma.1 in Appendix B, we have:

$$\begin{aligned} \dim(W_{n_w} \times \Phi_{n_\varphi} \setminus \ker \mathcal{Q}^{\Gamma_w} \mathcal{T}^{\Gamma_w}) &= \dim(W_{n_w}^{\Gamma} \setminus \ker \mathcal{Q}^{\Gamma_w}) \\ &= \dim W_{n_w}^{\Gamma} - \underbrace{\dim(\ker \mathcal{Q}^{\Gamma_w})}_0 \\ &= \dim W_{n_w}^{\Gamma} = n_w^{\Gamma} \end{aligned} \quad (67)$$

176 where, it can reveal from the expression of  $\mathcal{Q}^{\Gamma_w}$  in Eq.(58) that, the space  
 177  $\ker \mathcal{Q}^{\Gamma_w}$  is empty, i.e  $\dim(\ker \mathcal{Q}^{\Gamma_w}) = 0$ .

178 Consequently, substituting Eqs.(66) and (67) into Eq.(62), the right-hand  
 179 side of Eq.(60) can be rewritten as:

$$\begin{aligned} \max(n_w + 2n_\varphi - \min(n_w, n_s), n_w^{\Gamma}) &\leq \dim(W_{n_w} \times \Phi_{n_\varphi} \setminus \ker \mathcal{Q}) \\ &\leq n_w + 2n_\varphi - \min(n_w, n_s) + n_w^{\Gamma} \end{aligned} \quad (68)$$

180 It is clear that  $n_w^{\Gamma} \ll n_w + 2n_\varphi - \min(n_w, n_s)$  FIXME

$$n_q \leq n_\varphi \quad (69)$$

$$n_w^{\text{opt}} = \langle \lfloor n_q \rfloor - 1 \rangle, \quad n_q^{\text{opt}} = n_\varphi \quad (70)$$

182 **Remark 3.** It should be noted that, the proposed estimator is considered for  
 183 the entire domain and all the nodal discretization should be so well-posed that  
 184 the monomial basis number of equivalent polynomial space is equal to the degree  
 185 of freedoms for approximation space, i.g  $\dim W_{n_w} = \dim W_h = n_w$ . While  
 186 a local domain contains a too dense nodal distribution than others, the nodal  
 187 discretization satisfied the estimator will still meet the locking issue in local.

188 **Remark 4.** The shear force locking will influence the shear strain locking.

## 189 4. Shear-locking-free FE-Meshfree formulation

### 190 4.1. Reproducing kernel meshfree approximation

$$\Psi_I(\mathbf{x}) = \mathbf{c}^T(\mathbf{x}_I - \mathbf{x}) \mathbf{p}(\mathbf{x}_I - \mathbf{x}) \phi(\mathbf{x}_I - \mathbf{x}) \quad (71)$$

$$\phi(\mathbf{x}_I - \mathbf{x}) = \phi(r_{1I}) \phi(r_{2I}), \quad r_{iI} = \frac{|\mathbf{x}_{iI} - \mathbf{x}_i|}{s_i} \quad (72)$$

$$\phi(r) = \frac{1}{3!} \begin{cases} (2-2r)^3 - 4(1-2r)^3 & r \leq \frac{1}{2} \\ (2-2r)^3 & \frac{1}{2} < r \leq 1 \\ 0 & r > 1 \end{cases} \quad (73)$$

193

$$\sum_{I=1}^n \Psi_I(\mathbf{x}) \mathbf{p}(\mathbf{x}_I) = \mathbf{p}(\mathbf{x}) \quad (74)$$

194 or a shifted form

$$\sum_{I=1}^n \Psi_I(\mathbf{x}) \mathbf{p}(\mathbf{x}_I - \mathbf{x}) = \mathbf{p}(\mathbf{0}) \quad (75)$$

195

$$\mathbf{c}(\mathbf{x}_I - \mathbf{x}) = \mathbf{A}^{-1}(\mathbf{x}_I - \mathbf{x}) \mathbf{p}(\mathbf{0}) \quad (76)$$

196 with

$$\mathbf{A}(\mathbf{x}_I - \mathbf{x}) = \sum_{I=1}^n \mathbf{p}(\mathbf{x}_I - \mathbf{x}) \mathbf{p}^T(\mathbf{x}_I - \mathbf{x}) \phi(\mathbf{x}_I - \mathbf{x}) \quad (77)$$

197

$$\Psi_I(\mathbf{x}) = \mathbf{p}^T(\mathbf{0}) \mathbf{A}^{-1} \mathbf{p}(\mathbf{x}_I - \mathbf{x}) \phi(\mathbf{x}_I - \mathbf{x}) \quad (78)$$

198 *4.2. Eigenvalue test for checking LBB coefficients(should be fixed)*

199 To verify the optimal constraint ratios proposed in Eq. (70), the eigenvalue  
200 test of LBB coefficients [4] in a square domain, shown in Figure.??, is intro-  
201 duced herein, where all the boundaries of this domain are clamped. In order  
202 to adjust the constraint ratio more flexibly, the reproducing kernel meshfree  
203 approximation are used to discretize the deflection  $w$  and rotation  $\varphi$ , yields:

$$N_I^w := \Psi_I, N_K^\varphi := \Psi_K^\varphi, N_R^q := N_R \quad (79)$$

204 while, as shown in Figure.??, the meshfree nodes for  $w$  and  $\varphi$  are different.

205 To meet the variational consistency of the weak formulation with meshfree  
206 approximation, the reproducing kernel gradient smoothing integration frame-  
207 work (RKGSi) [5] is used herein. Accordingly, the Mindlin plate version of  
208 RKGSi framework is presented in Appendix C.

209 *4.3. Shear-locking-free formulations with optimal constraint ratio*

$$N_I^w := \Psi_I, N_K^\varphi := N_K, N_R^q := N_R \quad (80)$$

210 FIXME

211 **Remark 5.** *The discretization of Eq.*212 **5. Numerical examples**213 *5.1. Square plate subjected non-uniform transverse load*

214 Consider a square plate with the clamped boundary, as showing in Figure.  
215 , subjecting a non-uniform transverse load  $\bar{p}$ :

$$\bar{p} = \frac{Eh^3}{12(1-\nu^2)} \left( \begin{array}{l} 12y(y-1)(5x^2-5x+1)(2y^2(y-1)^2+x(x-1)(5y^2-5y+1)) \\ +12x(x-1)(5y^2-5y+1)(2x^2(x-1)^2+y(y-1)(5x^2-5x+1)) \end{array} \right) \quad (81)$$

216 In this circumstance, the exact solution of the deflection and rotation are listed  
 217 as follows:

$$w = \frac{1}{3}x^3(x-1)^3y^3(y-1)^3 - \frac{2h^2}{5(1-\nu)} \left( \frac{y^3(y-1)^3x(x-1)(5x^2-5x+1)+}{x^3(x-1)^3y(y-1)(5y^2-5y+1)} \right) \quad (82)$$

$$\begin{cases} \varphi_1 = y^3(y-1)^3x^2(x-1)^2(2x-1) \\ \varphi_2 = x^3(x-1)^3y^2(y-1)^2(2y-1) \end{cases} \quad (83)$$

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## 223 Appendix A. Complete-polynomial-based estimator for LBB coefficient 224

225 This appendix presents the detailed derivation of the LBB coefficient esti-  
226 mator used in this work. This estimator provides a quantitative measure of  
227 the stability of mixed finite element formulations just by counting the degree  
228 of freedoms (DOFs) of the approximation spaces. The main route of derivation  
229 follows the path proposed in Ref. [2]. Unlike the consideration of volumetric  
230 locking in [2], here we focus on the general LBB condition for any locking is-  
231 sue. Consider the following constraint servicing to a saddle-point problem: find  
232  $u \in U$  such that

$$b(p, u) = g(p), \quad \forall p \in P \quad (\text{A.1})$$

233 where  $b : U \times P \rightarrow \mathbb{R}$ ,  $g : P \rightarrow \mathbb{R}$  are bilinear and linear forms, respectively. For  
234 an any fixed  $p \in P$ , there exists a linear operator  $\mathcal{P} : U \rightarrow P$  satisfying:

$$b(p, u) = (p, \mathcal{P}u) \quad (\text{A.2})$$

235 where  $(\bullet, \bullet) : P \times P \rightarrow \mathbb{R}$  is the inner product in  $P$ .

236 By introducing the discrete spaces  $U_h \subset U$  and  $P_h \subset P$ , the constraint of  
237 Eq. (A.1) is approximated as: find  $u_h \in U_h$  such that

$$b(p_h, u_h) = g(p_h), \quad \forall p_h \in P_h \quad (\text{A.3})$$

238 and  $\mathcal{P}$ 's orthogonal projection operator regarding to  $P_h$ , namely  $\mathcal{P}_h : U_h \rightarrow P_h$ ,  
239 is defined by

$$(p_h, \mathcal{P}u_h - \mathcal{P}_h u_h) = 0 \Rightarrow (p_h, \mathcal{P}u_h) = (p_h, \mathcal{P}_h u_h), \quad \forall p_h \in P_h \quad (\text{A.4})$$

240 To obtain the satisfaction solution of saddle-point problems with constraint  
241 of Eq. (A.3), the well-known LBB condition is required, i.e., there exists a  
242 positive constant  $\beta > 0$ , namely LBB coefficient, such that

$$\inf_{p_h \in P_h} \sup_{u_h \in U_h} \frac{|b(p_h, u_h)|}{\|p\|_P \|u\|_U} \geq \beta \quad (\text{A.5})$$

243 where  $\|\bullet\|_U$  and  $\|\bullet\|_P$  are the norms in spaces  $U$  and  $P$ , respectively. To  
244 further estimate the LBB coefficient  $\beta$  by the degree of freedom of  $U_h$  and  $P_h$ ,  
245 i.e.  $n_u = \dim U_h$  and  $n_p = \dim P_h$ , we introduce the following theorem.

246 **Theorem 1.** (Completed-polynomial-based estimator for LBB coefficient) Sup-  
247 pose that  $\mathcal{P} : U \rightarrow P$  is the linear operator representing the bilinear form  $b$   
248 in LBB condition and  $\mathcal{P}_h : U_h \rightarrow P_h$  is the corresponding orthogonal projec-  
249 tion operator regarding to  $P_h$ ,  $U_{n_u} \subset U$  is a complete polynomial space with  
250  $\dim U_{n_u} = n_u$ . The LBB coefficient  $\beta$  can be estimated by:

$$\beta \leq \inf_{U' \subset U_{n_u} \setminus \ker \mathcal{P}_h \mathcal{I}_h} \sup_{u \in U'} \frac{\|\mathcal{P}u\|_P}{\|u\|_U} + O(h) \quad (\text{A.6})$$

251 where  $\mathcal{I}_h : U_{n_u} \rightarrow U_h$  is the interpolation operator with the characteristic size  
252 of the discretization  $h$ .

253 PROOF. Firstly, with the definition of  $\mathcal{P}$  and  $\mathcal{P}_h$  in Eqs. (A.2) and (A.4) re-  
 254 spectively, the LBB condition can be rewritten as:

$$\begin{aligned} \beta &\leq \inf_{p_h \in P_h} \sup_{u_h \in U_h} \frac{|(p_h, \mathcal{P}u_h)|}{\|p_h\|_P \|u_h\|_U} = \inf_{p_h \in P_h} \sup_{u_h \in U_h} \frac{|(p_h, \mathcal{P}_h u_h)|}{\|p_h\|_P \|u_h\|_U} \\ &\leq \inf_{p_h \in \text{Im} \mathcal{P}_h} \sup_{u \in U_h} \frac{|(p_h, \mathcal{P}_h u_h)|}{\|p_h\|_P \|u_h\|_U} \end{aligned} \quad (\text{A.7})$$

255 FIXME: the definition of image space can be canceled. where  $\text{Im} \mathcal{P}_h$  is the  
 256 image space of operator  $\mathcal{P}_h$ , i.e.  $\text{Im} \mathcal{P}_h := \{p_h \in P_h \mid \exists u_h \in U_h, \mathcal{P}_h u_h = p_h\}$ ,  
 257  $\text{Im} \mathcal{P}_h \subset P_h$ . Since both  $p_h$  and  $\mathcal{P}_h u_h$  belong to  $\text{Im} \mathcal{P}_h$ , using Cauchy-Schwarz  
 258 inequality, we can further obtain that:

$$|(\mathcal{P}_h u_h, p_h)| \leq \|\mathcal{P}_h u_h\|_P \|p_h\|_P \quad (\text{A.8})$$

259 For a given  $p_h$ , with introduction of a space  $U' := \{u_h \in U_h \mid \mathcal{P}_h u_h = p_h\}$   
 260 regarding to  $p_h$ , the equality in Eq. (A.7) holds when  $p_h = \mathcal{P}_h u_h$ , yields:

$$|(\mathcal{P}_h u_h, p_h)| = \|\mathcal{P}_h u_h\|_P \|p_h\|_P, \quad \forall u_h \in U' \quad (\text{A.9})$$

261 and, obviously,  $U' \subseteq U_h \setminus \ker \mathcal{P}_h$ ,  $\ker \mathcal{P}_h$  is the kernel space of operator  $\mathcal{P}_h$ , i.e.  
 262  $\ker \mathcal{P}_h := \{u_h \in U_h \mid \mathcal{P}_h u_h = 0\}$ . Substituting Eq. (A.9) into Eq. (A.7) leads  
 263 to:

$$\beta \leq \inf_{U' \subset U_h \setminus \ker \mathcal{P}_h} \sup_{u_h \in U'} \frac{\|\mathcal{P}_h u_h\|_P}{\|u_h\|_U} \quad (\text{A.10})$$

264 Secondly, the complete polynomial space  $V_{n_u} \subset U$  with identical dimension  
 265 to  $U_h$  is introduced herein to further estimate the  $\beta$ , where  $n_u$  is the dimension  
 266 of  $U_h$ , i.e.  $n_u = \dim U_h = \dim V_{n_u}$ . At this circumstance, for any  $u \in U_{n_u}$ ,  
 267 there exists an interpolation operator  $\mathcal{I}_h : U \rightarrow U_h$  such that  $u_h = \mathcal{I}_h u$ . Thus,  
 268 Eq. (A.10) can be rewritten as:

$$\beta \leq \inf_{U' \subset U_{n_u} \setminus \ker \mathcal{P}_h \mathcal{I}_h} \sup_{u \in U'} \frac{\|\mathcal{P}_h \mathcal{I}_h u\|_P}{\|\mathcal{I}_h u\|_U} \quad (\text{A.11})$$

269 In accordance with the triangular inequality [6] and the relationship of Eq.  
 270 (A.4), we have:

$$\begin{aligned} \|\mathcal{P}_h \mathcal{I}_h u\|_P &\leq \sup_{p_h \in P_h} \frac{|(\mathcal{P}_h \mathcal{I}_h u, p_h)|}{\|p_h\|_P} = \sup_{p_h \in P_h} \frac{|(\mathcal{P} \mathcal{I}_h u, p_h)|}{\|p_h\|_P} \\ &\leq \sup_{p_h \in P_h} \frac{|(\mathcal{P} u, p_h)| + |(\mathcal{P} u - \mathcal{P} \mathcal{I}_h u, p_h)|}{\|p_h\|_P} \\ &= \|\mathcal{P} u\|_P + \|\mathcal{P}(u - \mathcal{I}_h u)\|_P \end{aligned} \quad (\text{A.12})$$

271 where the last term is the interpolation error of space  $U_h$  with respect to operator  
 272  $\mathcal{P}$ . Generally, particularly for finite element approximations, the interpolation  
 273 error is proportional to the characteristic size of the discretization  $h$  [6]:

$$\|\mathcal{P}(u - \mathcal{I}_h u)\|_P \leq Ch^k |u|_{H_k} \quad (\text{A.13})$$

274 in which  $C$  is a positive constant independent of  $h$ ,  $k$  is the convergence order  
 275 related to the operators  $\mathcal{I}_h$  and  $\mathcal{P}$ . With the combination of Eqs. (A.11), (A.12)  
 276 and (A.13), the final estimator of LBB coefficient  $\beta$  in Eq. (1) can be proved:

$$\begin{aligned}
 \beta &\leq \inf_{U' \subset U_{n_u} \setminus \ker \mathcal{P}_h \mathcal{I}_h} \sup_{u \in U'} \frac{\|\mathcal{P}_h \mathcal{I}_h u\|_P}{\|\mathcal{I}_h u\|_U} \\
 &\leq \inf_{U' \subset U_{n_u} \setminus \ker \mathcal{P}_h \mathcal{I}_h} \sup_{u \in U'} \frac{\|\mathcal{P} u\|_P + \|\mathcal{P}(u - \mathcal{I}_h u)\|_P}{\|\mathcal{I}_h u\|_U} \\
 &\leq \inf_{U' \subset U_{n_u} \setminus \ker \mathcal{P}_h \mathcal{I}_h} \sup_{u \in U'} \frac{\|\mathcal{P} u\|_P}{\|u\|_U} + O(h)
 \end{aligned} \tag{A.14}$$

277        FIXME:use the remark env It should be noted that Theorem 1 inherently  
 278 embeds the numbers of DOFs of approximation spaces  $U_h$  and  $P_h$ , while  $n_u =$   
 279  $\dim U_h$  and  $n_p = \dim \text{Im} \mathcal{P}_h = \dim(U_h \setminus \ker \mathcal{P}_h \mathcal{I}_h)$ , thus it is easy to justify the  
 280 LBB condition by just counting the DOFs of approximation spaces.



281 **Appendix B. Preliminaries for functional analysis**

282 In the interest of conciseness, kernel space

$$\ker \mathcal{P} := \{u \in U \mid \mathcal{P}u = 0\} \quad (\text{B.1})$$

283 image space

$$\text{Im} \mathcal{P} := \{p \in P \mid \exists u \in U, p = \mathcal{P}u\} \quad (\text{B.2})$$

284 It is clear that  $\ker \mathcal{P} \subset U$  and  $\text{Im} \mathcal{P} \in P$ .

285 **FIXME:** Trace's theorem De Morgan's theorem Grassmann's theorem Rank-  
286 nullity theorem

287 **Lemma 1.** *Suppose that  $\mathcal{T} : U \rightarrow U|_{\Gamma}$  is the trace operator of  $U$ ,  $U|_{\Gamma}$  is  $U$ 's  
288 trace space on the boundary  $\Gamma$ . Let  $\mathcal{L}$  is an operator acting on  $U|_{\Gamma}$ , then we  
289 have:*

$$\dim(U \setminus \ker \mathcal{L}\mathcal{T}) = \dim(U|_{\Gamma} \setminus \ker \mathcal{L}) \quad (\text{B.3})$$

290 **PROOF.** In accordance with rank-nullity theorem, we have the following rela-  
291 tionship:

$$\dim U = \dim(\text{Im} \mathcal{L}\mathcal{T}) + \dim(\ker \mathcal{L}\mathcal{T}) \quad (\text{B.4})$$

292 or

$$\dim(U \setminus \ker \mathcal{L}\mathcal{T}) = \dim U - \dim(\ker \mathcal{L}\mathcal{T}) = \dim(\text{Im} \mathcal{L}\mathcal{T}) \quad (\text{B.5})$$

293 in which, according to the definition of the image space  $\text{Im} \mathcal{L}\mathcal{T}$ , it is clear that:

$$\text{Im} \mathcal{L}\mathcal{T} = \{\mathcal{L}\mathcal{T}u \mid u \in U\} = \{\mathcal{L}(\mathcal{T}u) \mid \mathcal{T}u \in U|_{\Gamma}\} = \text{Im} \mathcal{L} \quad (\text{B.6})$$

294 Subsequently, the dimension of image space for operator  $\mathcal{L}$ , i.e.  $\dim(\text{Im} \mathcal{L})$ ,  
295 can be evaluated by introducing rank-nullity theorem again as follows:

$$\dim U|_{\Gamma} = \dim(\text{Im} \mathcal{L}) + \dim(\ker \mathcal{L}) \quad (\text{B.7})$$

296 or

$$\dim(\text{Im} \mathcal{L}) = \dim U|_{\Gamma} - \dim(\ker \mathcal{L}) = \dim(U|_{\Gamma} \setminus \ker \mathcal{L}) \quad (\text{B.8})$$

297 With the combination of Eqs. (B.5), (B.6) and (B.8), the relationship of Eq.  
298 (B.3) can be proofed.

$$\begin{aligned}
\Pi(w, \boldsymbol{\varphi}, \mathbf{q}, \mathbf{m}) = & - \int_{\Omega} \frac{1}{2} \mathbf{m} : \mathbf{D}^{-1} : \mathbf{m} d\Omega - \int_{\Omega} \frac{1}{2} \frac{\mathbf{q} \cdot \mathbf{q}}{kGh} d\Omega \\
& - \int_{\Gamma_{\varphi}} \mathbf{n} \cdot \mathbf{m} \cdot \bar{\boldsymbol{\varphi}} d\Gamma - \int_{\Gamma_w} \mathbf{q} \cdot \mathbf{n} \bar{w} d\Gamma \\
& - \int_{\Gamma_m} \boldsymbol{\varphi} \cdot (\mathbf{m} \cdot \mathbf{n} - \bar{\mathbf{m}}_n) d\Gamma - \int_{\Omega} \boldsymbol{\varphi} \cdot (\nabla \cdot \mathbf{m} - \mathbf{q}) d\Omega \\
& + \int_{\Gamma_q} w(\mathbf{q} \cdot \mathbf{n} - \bar{\mathbf{q}}_n) d\Gamma - \int_{\Omega} w(\nabla \cdot \mathbf{q} + \bar{p}) d\Omega
\end{aligned} \tag{C.1}$$

$$\begin{aligned}
a_m(\delta \mathbf{m}, \mathbf{m}) + b_m(\delta \mathbf{m}, \boldsymbol{\varphi}) &= f_m(\delta \mathbf{m}), \quad \forall \delta \mathbf{m} \in M \\
a_q(\delta \mathbf{q}, \mathbf{q}) + b_q(\delta \mathbf{q}, \mathbf{u}) &= f_q(\delta \mathbf{q}), \quad \forall \delta \mathbf{q} \in Q \\
b_w(\delta w, \mathbf{q}) &= f_w(\delta w), \quad \forall \delta w \in W \\
b_m(\delta \boldsymbol{\varphi}, \mathbf{m}) + b_{\varphi}(\delta \boldsymbol{\varphi}, \mathbf{q}) &= f_{\varphi}(\delta \boldsymbol{\varphi}), \quad \forall \delta \boldsymbol{\varphi} \in \Phi
\end{aligned} \tag{C.2}$$

$$a_m(\delta \mathbf{m}, \mathbf{m}) = - \int_{\Omega} \delta \mathbf{m} : \frac{12}{t^3} \mathbf{D}^{-1} : \mathbf{m} d\Omega \tag{C.3}$$

$$b_m(\mathbf{m}, \boldsymbol{\varphi}) = - \int_{\Gamma_m} \mathbf{n} \cdot \mathbf{m} \cdot \boldsymbol{\varphi} d\Gamma + \int_{\Omega} \nabla \mathbf{m} \cdot \boldsymbol{\varphi} d\Omega \tag{C.4}$$

$$f_m(\boldsymbol{\varphi}, \mathbf{q}) = \int_{\Gamma_{\varphi}} \mathbf{n} \cdot \mathbf{m} \cdot \bar{\boldsymbol{\varphi}} d\Gamma \tag{C.5}$$

$$\tag{C.6}$$

$$\begin{aligned}
& - \int_{\Omega} \delta \mathbf{m} : \mathbf{D}^{-1} : \mathbf{m} d\Omega - \int_{\Gamma_m} \mathbf{n} \cdot \delta \mathbf{m} \cdot \boldsymbol{\varphi} d\Gamma + \int_{\Omega} \nabla \delta \mathbf{m} \cdot \boldsymbol{\varphi} d\Omega = \int_{\Gamma_{\varphi}} \mathbf{n} \cdot \delta \mathbf{m} \cdot \bar{\boldsymbol{\varphi}} d\Gamma \\
& \tag{C.7}
\end{aligned}$$

$$\begin{aligned}
& - \int_{\Omega} \frac{\delta \mathbf{q} \cdot \mathbf{q}}{kGh} d\Omega + \int_{\Gamma_q} \delta \mathbf{q} \cdot \mathbf{n} w d\Gamma - \int_{\Omega} \nabla \cdot \delta \mathbf{q} w d\Omega + \int_{\Omega} \delta \mathbf{q} \cdot \boldsymbol{\varphi} d\Omega = - \int_{\Gamma_w} \delta \mathbf{q} \cdot \mathbf{n} \bar{w} d\Gamma \\
& \tag{C.8}
\end{aligned}$$

$$\begin{aligned}
& \int_{\Gamma_q} \delta w \mathbf{q} \cdot \mathbf{n} d\Gamma - \int_{\Omega} \delta w \nabla \cdot \mathbf{q} d\Omega = \int_{\Gamma_q} \delta w \bar{\mathbf{q}}_n d\Gamma - \int_{\Omega} \delta w \bar{p} d\Omega \\
& \tag{C.9}
\end{aligned}$$

$$\begin{aligned}
& - \int_{\Gamma_m} \delta \boldsymbol{\varphi} \cdot \mathbf{m} \cdot \mathbf{n} d\Gamma - \int_{\Omega} \delta \boldsymbol{\varphi} \cdot \mathbf{m} \cdot \nabla d\Omega + \int_{\Omega} \delta \boldsymbol{\varphi} \cdot \mathbf{q} d\Omega = - \int_{\Gamma_m} \delta \boldsymbol{\varphi} \cdot \bar{\mathbf{m}}_n d\Gamma \\
& \tag{C.10}
\end{aligned}$$

$$\begin{aligned}
& - \int_{\Omega_C} \delta m_{\alpha\beta} D_{\alpha\beta\gamma\eta}^{-1} m_{\gamma\eta} d\Omega - \int_{\Gamma_C} \delta M_{\alpha\beta} n_{\beta} \varphi_{\alpha} d\Gamma + \int_{\Omega_C} \delta M_{\alpha\beta, \beta} \varphi_{\alpha} d\Omega + \int_{\Gamma_{\varphi} \cap \Gamma_C} \delta M_{\alpha\beta} n_{\beta} \varphi_{\alpha} d\Gamma = \int_{\Gamma_{\varphi} \cap \Gamma_C} M_{\alpha\beta} n_{\beta} \bar{\varphi}_{\alpha} d\Gamma \\
& \tag{C.11}
\end{aligned}$$

$$m_{\alpha\beta h}(\mathbf{x}) = \mathbf{p}^T(\mathbf{x}) \mathbf{a}_{\alpha\beta C} \quad \text{in } \Omega_C \quad (\text{C.12})$$

$$\mathbf{a}_{\alpha\beta C} = D_{\alpha\beta\eta\gamma} \mathbf{G}_C^{-1} \left( \sum_{K=1}^{n_\varphi} (\tilde{\mathbf{g}}_{\eta K} + \hat{\mathbf{g}}_{\eta K}) \varphi_{\gamma K} + \bar{\mathbf{g}}_{\gamma\eta} \right) \quad (\text{C.13})$$

$$\tilde{\mathbf{g}}_{\alpha K} = - \int_{\Gamma_C} \mathbf{p} N_K^\varphi n_\alpha d\Gamma + \int_{\Omega_C} \mathbf{p}_{,\alpha} N_K^\varphi d\Omega \quad (\text{C.14})$$

$$\bar{\mathbf{g}}_{\alpha K} = \int_{\Gamma_C \cap \Gamma_\varphi} \mathbf{p} N_K^\varphi n_\alpha d\Gamma \quad (\text{C.15})$$

$$\hat{\mathbf{g}}_{\alpha\beta} = \int_{\Gamma_C \cap \Gamma_\varphi} \mathbf{p} n_\beta \bar{\varphi}_\alpha d\Gamma \quad (\text{C.16})$$

$$(\text{C.17})$$

$$\mathbf{m}_h = \tilde{\mathbf{m}}_h + \hat{\mathbf{m}}_h + \bar{\mathbf{m}}_h \quad (\text{C.18})$$

$$\mathbf{m}(\varphi_h) = \mathbf{D} : (\tilde{\boldsymbol{\kappa}}(\varphi_h) + \hat{\boldsymbol{\kappa}}_h + \bar{\boldsymbol{\kappa}}_{\gamma\eta}) \quad (\text{C.19})$$

$$\tilde{m}_{\alpha\beta h} = -D_{\alpha\beta\eta\gamma} (\tilde{\kappa}_{\gamma\eta h} + \bar{\kappa}_{\gamma\eta} + \hat{\kappa}_{\gamma\eta}) \quad (\text{C.20})$$

with

$$\tilde{\kappa}_{\alpha\beta h}(\mathbf{x}) = -\frac{1}{2} \sum_{K=1}^{n_\varphi} \left( \tilde{N}_{K,\alpha}^\varphi(\mathbf{x}) \varphi_{\beta K} + \tilde{N}_{K,\beta}^\varphi(\mathbf{x}) \varphi_{\alpha K} \right) \quad (\text{C.21})$$

$$\bar{\kappa}_{\alpha\beta h}(\mathbf{x}) = -\frac{1}{2} \sum_{K=1}^{n_\varphi} \left( \bar{N}_{K,\alpha}^\varphi(\mathbf{x}) \varphi_{\beta K} + \bar{N}_{K,\beta}^\varphi(\mathbf{x}) \varphi_{\alpha K} \right) \quad (\text{C.22})$$

$$\hat{\kappa}_{\alpha\beta h}(\mathbf{x}) = -\mathbf{G}_C^{-1} \hat{\mathbf{g}}_{\alpha\beta} \quad (\text{C.23})$$

$$\tilde{N}_{K,\alpha}^\varphi(\mathbf{x}) = \mathbf{p}^T(\mathbf{x}) \mathbf{G}_C^{-1} \tilde{\mathbf{g}}_{\alpha K} \quad (\text{C.24})$$

$$\bar{N}_{K,\alpha}^\varphi(\mathbf{x}) = \mathbf{p}^T(\mathbf{x}) \mathbf{G}_C^{-1} \bar{\mathbf{g}}_{\alpha K} \quad (\text{C.25})$$

$$\begin{aligned} & b_m(\delta\varphi_h, \mathbf{m}_h) \\ &= - \int_{\Gamma_m} \delta\varphi_h \cdot \mathbf{m}_h \cdot \mathbf{n} d\Gamma + \int_{\Omega} \delta\varphi_h \cdot \mathbf{m}_h \cdot \nabla d\Omega \\ &= - \int_{\Gamma} \delta\varphi_{\alpha h} m_{\alpha\beta h} n_\beta d\Gamma + \int_{\Omega} \delta m_{\alpha\beta h, \beta} \varphi_{\alpha h} d\Omega + \int_{\Gamma_\varphi} \delta\varphi_{\alpha h} m_{\alpha\beta h} n_\beta d\Gamma \\ &= \sum_{C=1}^{n_e} \sum_{K,L=1}^{n_\varphi} \delta\varphi_{\alpha K} \left( - \int_{\Gamma_C} \mathbf{p} N_K^\varphi n_\beta d\Gamma + \int_{\Omega_C} \mathbf{p}_{,\beta} N_K^\varphi d\Omega + \int_{\Gamma_C \cap \Gamma_\varphi} \mathbf{p} N_K^\varphi n_\beta d\Gamma \right) D_{\alpha\beta\gamma\eta} \mathbf{G}_C^{-1} ((\tilde{\mathbf{g}}_{\eta L} + \bar{\mathbf{g}}_{\eta L}) \varphi_{\gamma L} + \hat{\mathbf{g}}_{\gamma\eta}) \\ &= \sum_{C=1}^{n_e} \sum_{K,L=1}^{n_\varphi} \delta\varphi_{\alpha K} \left( \int_{\Omega_C} \tilde{\mathbf{g}}_{\beta K}^T \mathbf{p} D_{\alpha\beta\gamma\eta} \mathbf{G}_C^{-1} \tilde{\mathbf{g}}_{\eta L} \right) \varphi_{\gamma L} \end{aligned} \quad (\text{C.26})$$

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$$\begin{aligned}
& b_m(\delta\varphi_h, \tilde{\mathbf{m}}_h) \\
&= - \int_{\Gamma_m} \delta\varphi_h \cdot \tilde{\mathbf{m}}_h \cdot \mathbf{n} d\Gamma + \int_{\Omega} \delta\varphi_h \cdot \tilde{\mathbf{m}}_h \cdot \nabla d\Omega \\
&= - \underbrace{\int_{\Gamma} \delta\varphi_h \cdot \tilde{\mathbf{m}}_h \cdot \mathbf{n} d\Gamma + \int_{\Omega} \delta\varphi_h \cdot \tilde{\mathbf{m}}_h \cdot \nabla d\Omega}_{\int_{\Omega} \delta\tilde{\kappa} : \frac{t^3}{12} \mathbf{D} : \tilde{\kappa} d\Omega} + \int_{\Gamma_{\varphi}} \delta\varphi_h \cdot \tilde{\mathbf{m}}_h \cdot \mathbf{n} d\Gamma \quad (\text{C.27})
\end{aligned}$$

315

$$b_m(\delta\varphi_h, \hat{\mathbf{m}}_h) = \tilde{b}_{\varphi}(\varphi_h, \tilde{\mathbf{m}}(\delta\varphi_h)) + \tilde{b}_{\varphi}(\delta\varphi_h, \hat{\mathbf{m}}(\varphi_h)) \quad (\text{C.28})$$

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$$b_m(\delta\varphi_h, \bar{\mathbf{m}}_h) = \hat{f} + \bar{f} \quad (\text{C.29})$$

317

$$\begin{aligned}
& b_m(\delta\varphi_h, \tilde{\mathbf{m}}_h) \\
&= - \int_{\Gamma_m} \delta\varphi_h \cdot \tilde{\mathbf{m}}_h \cdot \mathbf{n} d\Gamma + \int_{\Omega} \delta\varphi_h \cdot \tilde{\mathbf{m}}_h \cdot \nabla d\Omega \\
&= - \int_{\Gamma} \delta\varphi_{\alpha h} \tilde{m}_{\alpha\beta h} n_{\beta} d\Gamma + \int_{\Omega} \delta\tilde{m}_{\alpha\beta h, \beta} \varphi_{\alpha h} d\Omega + \int_{\Gamma_{\varphi}} \delta\varphi_{\alpha h} \tilde{m}_{\alpha\beta h} n_{\beta} d\Gamma \\
&= \sum_{C=1}^{n_e} \sum_{K,L=1}^{n_{\varphi}} \delta\varphi_{\alpha K} \left( - \int_{\Gamma_C} \mathbf{p} N_K^{\varphi} n_{\beta} d\Gamma + \int_{\Omega_C} \mathbf{p}_{,\beta} N_K^{\varphi} d\Omega + \int_{\Gamma_C \cap \Gamma_{\varphi}} \mathbf{p} N_K^{\varphi} n_{\beta} d\Gamma \right) D_{\alpha\beta\gamma\eta} \mathbf{G}_C^{-1} \tilde{\mathbf{g}}_{\eta L} \\
&= \sum_{C=1}^{n_e} \sum_{K,L=1}^{n_{\varphi}} \delta\varphi_{\alpha K} (\tilde{\mathbf{g}}_{\beta K} + \hat{\mathbf{g}}_{\beta K}) \mathbf{G}_C^{-1} \int_{\Omega_C} \mathbf{p} D_{\alpha\beta\gamma\eta} \mathbf{p}^T d\Omega \mathbf{G}_C^{-1} \tilde{\mathbf{g}}_{\eta L} \\
&= \sum_{C=1}^{n_e} \left( \underbrace{\int_{\Omega_C} \sum_{K=1}^{n_{\varphi}} \mathbf{p}^T \mathbf{G}_C^{-1} \tilde{\mathbf{g}}_{\beta K} \delta\varphi_{\alpha K} D_{\alpha\beta\gamma\eta}}_{\delta\tilde{\kappa}_{\alpha\beta}} \underbrace{\sum_{L=1}^{n_{\varphi}} \mathbf{G}_C^{-1} \tilde{\mathbf{g}}_{\eta L} \varphi_{\gamma L}}_{\tilde{\kappa}_{\gamma\eta}} \right) \\
&+ \sum_{C=1}^{n_e} \left( \underbrace{\int_{\Omega_C} \sum_{K=1}^{n_{\varphi}} \mathbf{p}^T \mathbf{G}_C^{-1} \bar{\mathbf{g}}_{\beta K} \delta\varphi_{\alpha K} D_{\alpha\beta\gamma\eta}}_{\delta\bar{\kappa}_{\alpha\beta}} \underbrace{\sum_{L=1}^{n_{\varphi}} \mathbf{G}_C^{-1} \tilde{\mathbf{g}}_{\eta L} \varphi_{\gamma L}}_{\tilde{\kappa}_{\gamma\eta}} \right) \\
&= a_{\varphi}(\tilde{\kappa}(\delta\varphi), \tilde{\kappa}(\varphi)) + a_{\varphi}(\hat{\kappa}(\delta\varphi), \tilde{\kappa}(\varphi)) \quad (\text{C.30})
\end{aligned}$$

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